DBMS Lab Assignment-1 — Solutions

Q1: Display the data in existing default employees or emp or employee table.

```
SELECT * FROM employees;
-- OR: SELECT * FROM emp;
-- OR: SELECT * FROM employee;
```

Output:

```
Note: Output depends on the pre-existing table in your schema (default Oracle
sample table EMP is shown here):
```

```
EMPNO ENAME      JOB       MGR HIREDATE   SAL  COMM DEPTNO
----- ---------- --------- --- --------- ---- ----- ------
 7369 SMITH      CLERK    7902 17-DEC-80  800          20
 7499 ALLEN      SALESMAN 7698 20-FEB-81 1600   300    30
 7521 WARD       SALESMAN 7698 22-FEB-81 1250   500    30
 7566 JONES      MANAGER  7839 02-APR-81 2975          20
 7654 MARTIN     SALESMAN 7698 28-SEP-81 1250  1400    30
 7698 BLAKE      MANAGER  7839 01-MAY-81 2850          30
 7782 CLARK      MANAGER  7839 09-JUN-81 2450          10
 7788 SCOTT      ANALYST  7566 09-DEC-82 3000          20
 7839 KING       PRESIDENT     17-NOV-81 5000          10
 7844 TURNER     SALESMAN 7698 08-SEP-81 1500     0    30
 7876 ADAMS      CLERK    7788 12-JAN-83 1100          20
 7900 JAMES      CLERK    7698 03-DEC-81  950          30
 7902 FORD       ANALYST  7566 03-DEC-81 3000          20
 7934 MILLER     CLERK    7782 23-JAN-82 1300          10
14 rows selected.
```

Q2: Display the structure of the above table.

```
DESC employees;
-- OR: DESCRIBE employees;
```

Output:

```
Name      Null? Type
--------- ----- -------------
EMPNO     NOT NULL NUMBER(4)
```

```
ENAME              VARCHAR2(10)
JOB                VARCHAR2(9)
MGR                NUMBER(4)
HIREDATE           DATE
SAL                NUMBER(7,2)
COMM               NUMBER(7,2)
DEPTNO             NUMBER(2)
```

Q3: Create a table "Student" with the following structure:
Roll Number(5), Name Varchar2(30), Age Number(5), Course Varchar2(5),
Math Number(6,2), Physics Number(6,2), Computer Number(6,2), Birthday Date.

```
CREATE TABLE Student (
    Roll     NUMBER(5),
    Name     VARCHAR2(30),
    Age      NUMBER(5),
    Course   VARCHAR2(5),
    Math     NUMBER(6, 2),
    Physics  NUMBER(6, 2),
    Computer NUMBER(6, 2),
    Birthday DATE
);
```

Output:

```
Table created.
```

```
DESC Student;
Name     Null? Type
-------- ----- -----------
ROLL            NUMBER(5)
NAME            VARCHAR2(30)
AGE             NUMBER(5)
COURSE          VARCHAR2(5)
MATH            NUMBER(6,2)
PHYSICS         NUMBER(6,2)
COMPUTER        NUMBER(6,2)
BIRTHDAY        DATE
```

Q4: Create table MSc from the Student table with the same fields and same structure but without any data.

```
CREATE TABLE MSc AS SELECT * FROM Student WHERE 1 = 0;
```

Output:

```
Table created.
```

Q5: Display the structure of MSc table.

```
DESC MSc;
-- OR: DESCRIBE MSc;
```

Output:

```
DESC MSc;
Name     Null? Type
-------- ----- -----------
ROLL            NUMBER(5)
NAME            VARCHAR2(30)
AGE             NUMBER(5)
COURSE          VARCHAR2(5)
MATH            NUMBER(6,2)
PHYSICS         NUMBER(6,2)
COMPUTER        NUMBER(6,2)
BIRTHDAY        DATE
```

Q6: Create table MCA from the Student table with the same fields and same structure but without any data. Rename Course → Department and Name → First_Name.

```
CREATE TABLE MCA AS
SELECT Roll, Name AS First_Name, Age,
       Course AS Department, Math, Physics, Computer, Birthday
FROM Student WHERE 1 = 0;
```

Output:

```
Table created.
```

Q7: Display the structure of MCA table.

```
DESC MCA;
-- OR: DESCRIBE MCA;
```

Output:

```
DESC MCA;
Name       Null? Type
---------- ----- -----------
```

```
ROLL             NUMBER(5)
FIRST_NAME       VARCHAR2(30)
AGE              NUMBER(5)
DEPARTMENT       VARCHAR2(5)
MATH             NUMBER(6,2)
PHYSICS          NUMBER(6,2)
COMPUTER         NUMBER(6,2)
BIRTHDAY         DATE
```

Q8: Insert the following records into the Student table:

```
INSERT INTO Student VALUES (1, 'Rahul',   19, 'BCA', 79.5, 67, 89,
'15-jun-93');
INSERT INTO Student VALUES (2, 'Kunal',   21, 'BCA', 68,   76, 59.5,
'16-aug-91');
INSERT INTO Student VALUES (3, 'Aditi',   20, 'MSc', 90,   73, 56,
'20-sep-92');
INSERT INTO Student VALUES (4, 'Sumit',   20, 'MCA', 57.5, 78, 81,
'07-dec-91');
INSERT INTO Student VALUES (5, 'Anirban', 22, 'MCA', 80,   68, 63,
'15-sep-94');
INSERT INTO Student VALUES (6, 'Kumkum',  21, 'BCA', 72,   54.5, 60,
'08-feb-95');
INSERT INTO Student VALUES (7, 'Suman',   21, 'BCA', 91.5, 32, 61,
'10-mar-94');
INSERT INTO Student VALUES (8, 'Rohit',   22, 'MSc', 85,   76, 92,
'19-apr-92');
```

Output:

```
1 row created.
1 row created.
1 row created.
1 row created.
1 row created.
1 row created.
1 row created.
1 row created.
```

Q9: Display all the students details from Student table.

```
SELECT * FROM Student;
```

Output:

```
ROLL NAME     AGE COURSE MATH PHYSICS COMPUTER BIRTHDAY
```

```
---- -------- --- ------ ---- ------- -------- --------
   1 Rahul     19 BCA    79.5      67       89 15-JUN-93
   2 Kunal     21 BCA      68      76     59.5 16-AUG-91
   3 Aditi     20 MSc      90      73       56 20-SEP-92
   4 Sumit     20 MCA    57.5      78       81 07-DEC-91
   5 Anirban   22 MCA      80      68       63 15-SEP-94
   6 Kumkum    21 BCA      72    54.5       60 08-FEB-95
   7 Suman     21 BCA    91.5      32       61 10-MAR-94
   8 Rohit     22 MSc      85      76       92 19-APR-92
8 rows selected.
```

Q10: Find out the details of the student with Roll no 5 from Student table.

```
SELECT * FROM Student WHERE Roll = 5;
```

Output:

```
ROLL NAME     AGE COURSE MATH PHYSICS COMPUTER BIRTHDAY
---- -------- --- ------ ---- ------- -------- --------
   5 Anirban   22 MCA      80      68       63 15-SEP-94
1 row selected.
```

Q11: Show the Roll, Name, marks of all subjects for all students.

```
SELECT Roll, Name, Math, Physics, Computer FROM Student;
```

Output:

```
ROLL NAME     MATH PHYSICS COMPUTER
---- -------- ---- ------- --------
   1 Rahul    79.5      67       89
   2 Kunal      68      76     59.5
   3 Aditi      90      73       56
   4 Sumit    57.5      78       81
   5 Anirban    80      68       63
   6 Kumkum     72    54.5       60
   7 Suman    91.5      32       61
   8 Rohit      85      76       92
8 rows selected.
```

Q12: Display all the student details belonging to the BCA department.

```
SELECT * FROM Student WHERE Course = 'BCA';
```

Output:

```
ROLL NAME     AGE COURSE MATH PHYSICS COMPUTER BIRTHDAY
---- -------- --- ------ ---- ------- -------- --------
   1 Rahul     19 BCA    79.5      67       89 15-JUN-93
   2 Kunal     21 BCA      68      76     59.5 16-AUG-91
   6 Kumkum    21 BCA      72    54.5       60 08-FEB-95
   7 Suman     21 BCA    91.5      32       61 10-MAR-94
4 rows selected.
```

Q13: Insert data in the MCA table from Student table where course is MCA.
(Note: MCA table has renamed columns — Name→First_Name, Course→Department)

```
INSERT INTO MCA (Roll, First_Name, Age, Department, Math, Physics,
Computer, Birthday)
SELECT Roll, Name, Age, Course, Math, Physics, Computer, Birthday
FROM Student WHERE Course = 'MCA';
```

Output:

```
2 rows created.
```

```
SELECT * FROM MCA;
```

```
ROLL FIRST_NAME AGE DEPARTMENT MATH PHYSICS COMPUTER BIRTHDAY
---- ---------- --- ---------- ---- ------- -------- --------
   4 Sumit       20 MCA        57.5      78       81 07-DEC-91
   5 Anirban     22 MCA          80      68       63 15-SEP-94
2 rows selected.
```

Q14: Insert data in the MSc table from Student table where course is MSc.

```
INSERT INTO MSc SELECT * FROM Student WHERE Course = 'MSc';
```

Output:

```
2 rows created.
```

```
SELECT * FROM MSc;
```

```
ROLL NAME     AGE COURSE MATH PHYSICS COMPUTER BIRTHDAY
---- -------- --- ------ ---- ------- -------- --------
   3 Aditi     20 MSc      90      73       56 20-SEP-92
   8 Rohit     22 MSc      85      76       92 19-APR-92
2 rows selected.
```

Q15: Display the structure of Student table.

```
DESC Student;
-- OR: DESCRIBE Student;
```

Output:

```
DESC Student;
Name     Null? Type
-------- ----- -----------
ROLL           NUMBER(5)
NAME           VARCHAR2(30)
AGE            NUMBER(5)
COURSE         VARCHAR2(5)
MATH           NUMBER(6,2)
PHYSICS        NUMBER(6,2)
COMPUTER       NUMBER(6,2)
BIRTHDAY       DATE
```

Q16: Display all the details of each student in Student table with Course appearing first in the display column list.

```
SELECT Course, Roll, Name, Age, Math, Physics, Computer, Birthday FROM
Student;
```

Output:

```
COURSE ROLL NAME     AGE MATH PHYSICS COMPUTER BIRTHDAY
------ ---- -------- --- ---- ------- -------- --------
BCA       1 Rahul     19 79.5      67       89 15-JUN-93
BCA       2 Kunal     21   68      76     59.5 16-AUG-91
MSc       3 Aditi     20   90      73       56 20-SEP-92
MCA       4 Sumit     20 57.5      78       81 07-DEC-91
MCA       5 Anirban   22   80      68       63 15-SEP-94
BCA       6 Kumkum    21   72    54.5       60 08-FEB-95
BCA       7 Suman     21 91.5      32       61 10-MAR-94
MSc       8 Rohit     22   85      76       92 19-APR-92
8 rows selected.
```

Q17: Update the Math marks of the student with Roll no 7 from 91.5 to 95 in the Student table.

```
UPDATE Student SET Math = 95 WHERE Roll = 7;
```

Output:

```
1 row updated.
```

```
SELECT * FROM Student WHERE Roll = 7;
```

```
ROLL NAME     AGE COURSE MATH PHYSICS COMPUTER BIRTHDAY
---- -------- --- ------ ---- ------- -------- --------
   7 Suman     21 BCA      95      32       61 10-MAR-94
1 row selected.
```

Q18: Update the Name of the student with Roll no 4 from Sumit to Sumitava in the Student table.

```
UPDATE Student SET Name = 'Sumitava' WHERE Roll = 4;
```

Output:

```
1 row updated.
```

```
SELECT * FROM Student WHERE Roll = 4;
```

```
ROLL NAME     AGE COURSE MATH PHYSICS COMPUTER BIRTHDAY
---- -------- --- ------ ---- ------- -------- --------
   4 Sumitava  20 MCA    57.5      78       81 07-DEC-91
1 row selected.
```

Q19: Delete the details of the student with Roll no 2 from the Student table.

```
DELETE FROM Student WHERE Roll = 2;
```

Output:

```
1 row deleted.
```

```
SELECT * FROM Student;
```

```
ROLL NAME     AGE COURSE MATH PHYSICS COMPUTER BIRTHDAY
---- -------- --- ------ ---- ------- -------- --------
   1 Rahul     19 BCA    79.5      67       89 15-JUN-93
   3 Aditi     20 MSc      90      73       56 20-SEP-92
```

```
   4 Sumitava  20 MCA    57.5       78        81 07-DEC-91
   5 Anirban   22 MCA      80       68        63 15-SEP-94
   6 Kumkum    21 BCA      72     54.5        60 08-FEB-95
   7 Suman     21 BCA      95       32        61 10-MAR-94
   8 Rohit     22 MSc      85       76        92 19-APR-92
7 rows selected.
```

Q20: Delete the details of all students from the Student table.

```
DELETE FROM Student;
-- OR: TRUNCATE TABLE Student;
```

Output:

```
Note: run after Q19, 7 rows remain in the table.


DELETE FROM Student;
7 rows deleted.


-- OR --


TRUNCATE TABLE Student;
Table truncated.


SELECT * FROM Student;
no rows selected
```